Graphic Systems

Lecture 2

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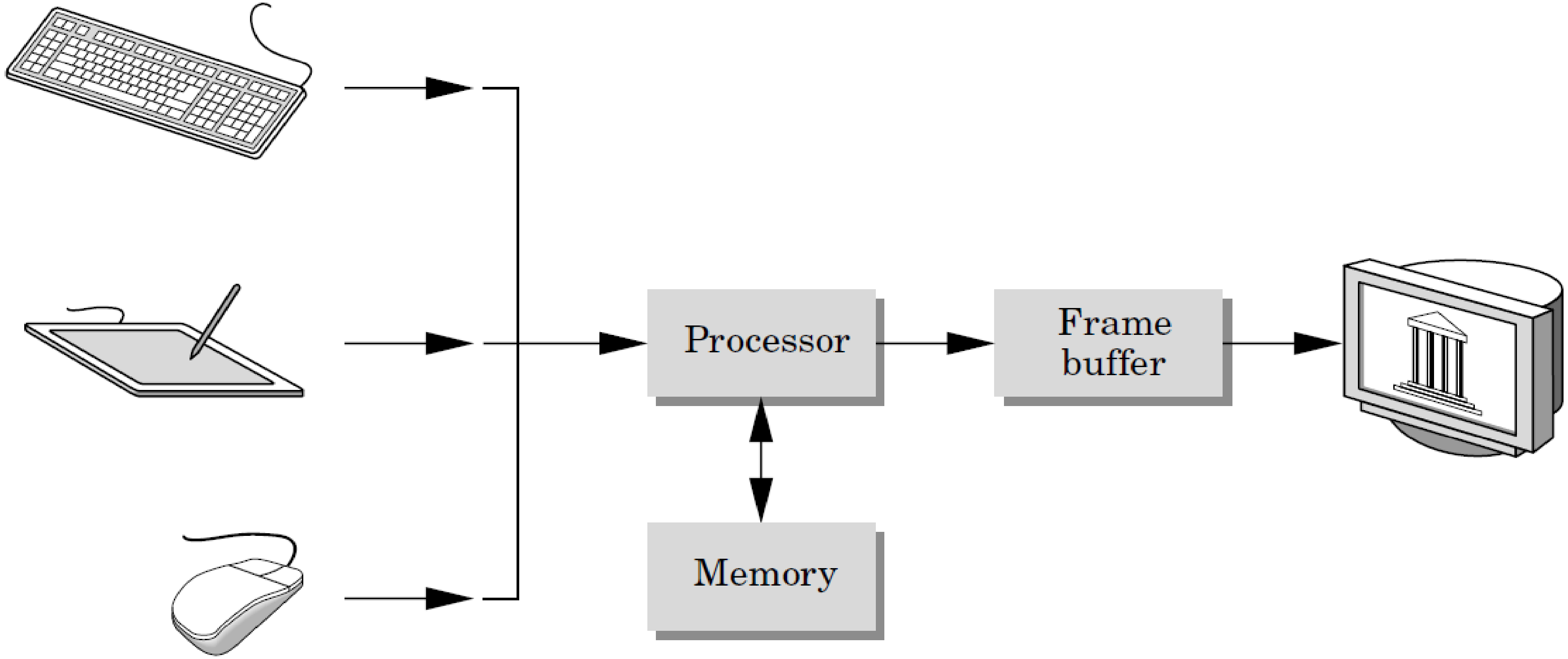
Outline

- Graphic input devices.
- Graphic Processing.
  - Framebuffer
- Output devices.

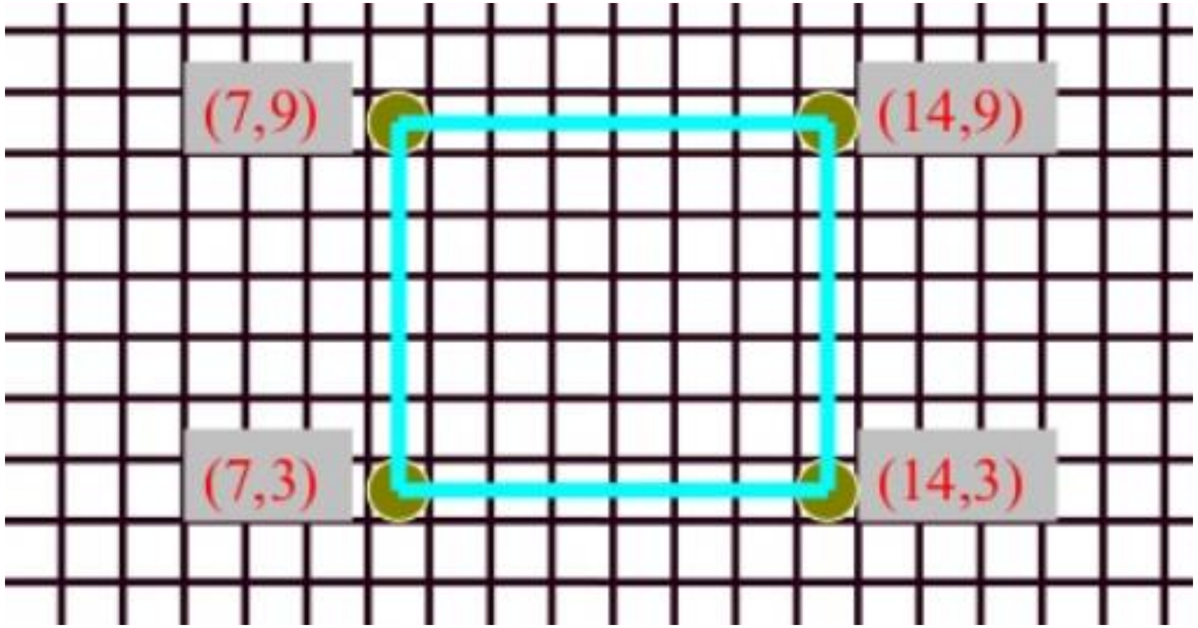


Input → Computation → Output

- When you want to draw a rectangle, How do we describe it to the computer ?

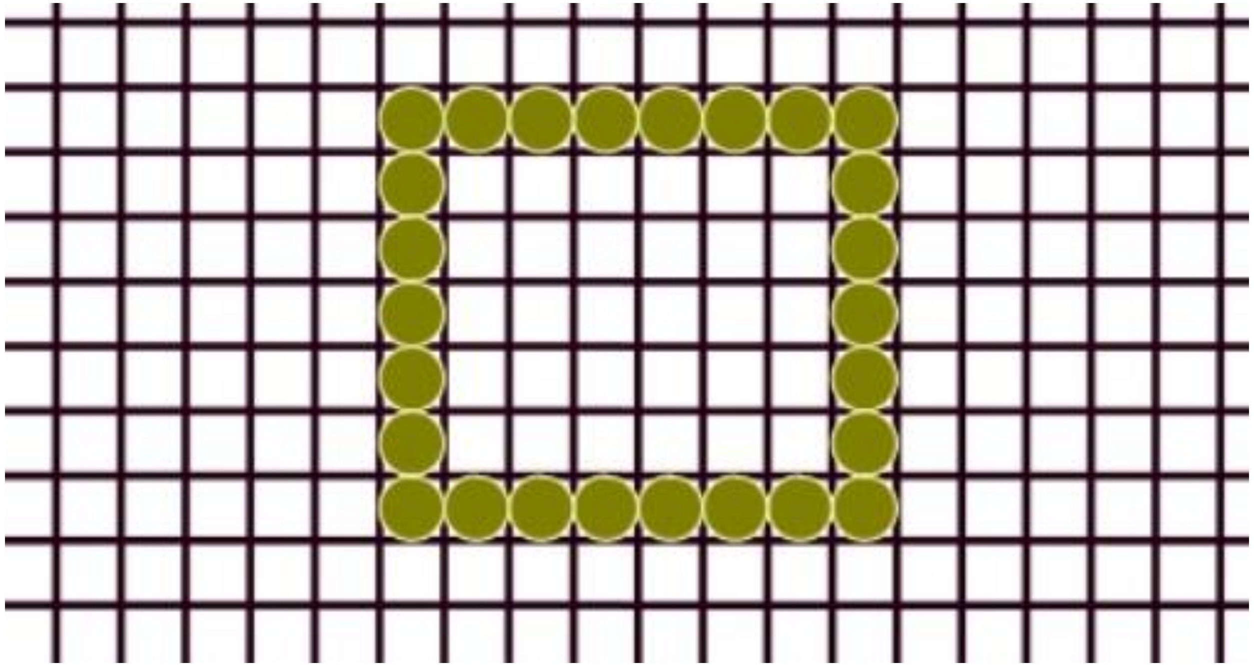
Model (n) – The Object Description that a Computer understands

Partition the Space



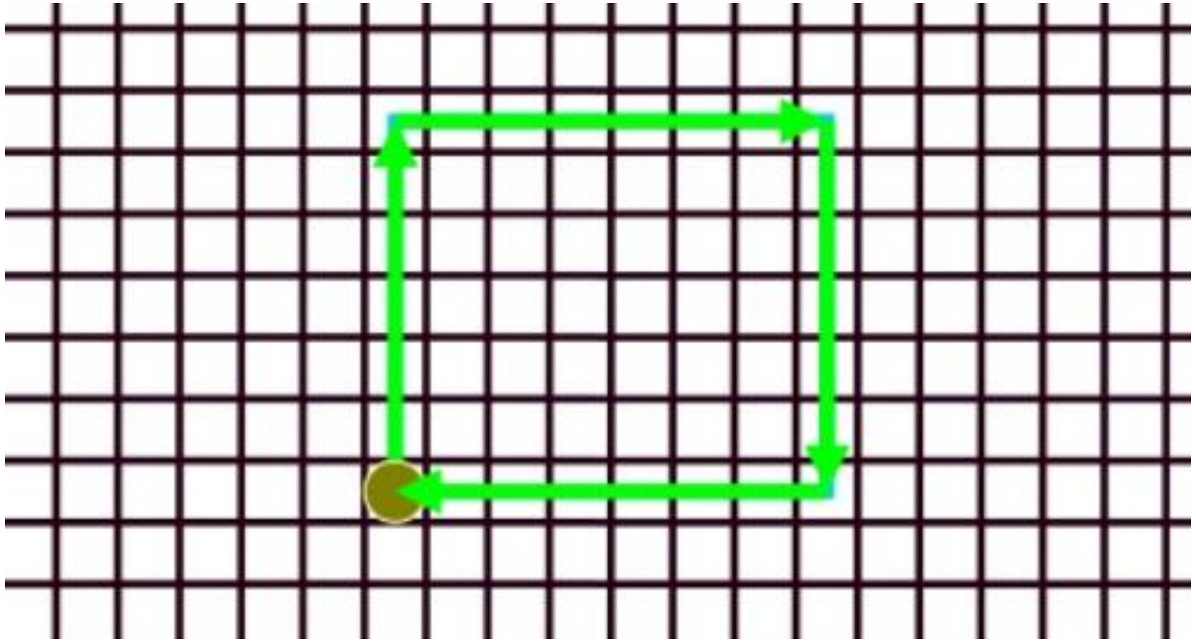
- Define a set of points (vertices) in 2D space.
- Given a set of vertices, draw lines between consecutive vertices.

Record Every Position



- Bitmap – A rectangular array of bits mapped one-to-one with pixels.

Position Relative



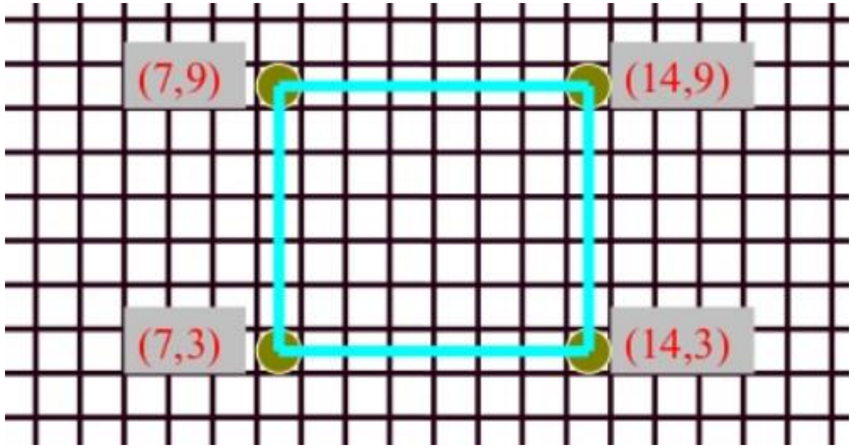
What are the properties needed to draw the rectangle ?

- x, y
- Width , height
- Fill
- Stroke (color), stroke width

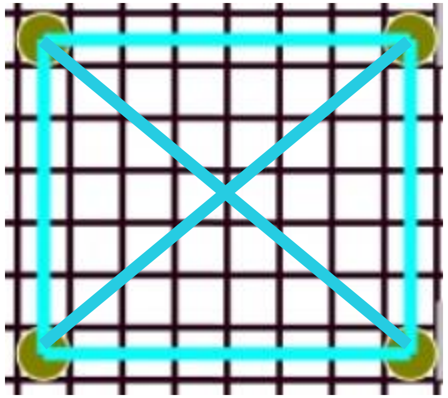
- Vector display system - graphical output system that was based on strokes (as opposed to pixels). Also known as: random, calligraphic, or stroke displays.

Model File for Rectangle

- Vertex Method is the most common method of representing objects
  - v 4 e 4
  - 7 3
  - 7 9
  - 14 9
  - 14 3
  - 1 2
  - 2 3
  - 3 4
  - 4 1



- Why Connectivity information Important ?



Input Devices

- Locator Devices
- Keyboard
- Scanner
  - Images
  - Laser
- Light Pens
- Voice Systems
- Touch Panels
- Camera/Vision Based

Locator Devices

- When queried, locator devices return a position and/or orientation
  - Mouse (2D and 3D)
  - Trackball
  - Joystick (2D and 3D)
  - Tablet
  - Virtual Reality Trackers
    - Data Gloves
    - Digitizers

Keyboard

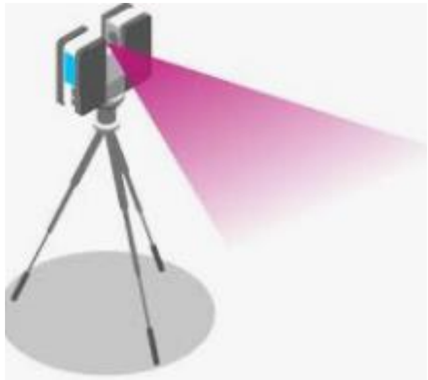
- Text input
  - List boxes, GUI
  - CAD/CAM
  - Modeling

- Hard coded
  - Vertex locations are inserted into code

Scanner

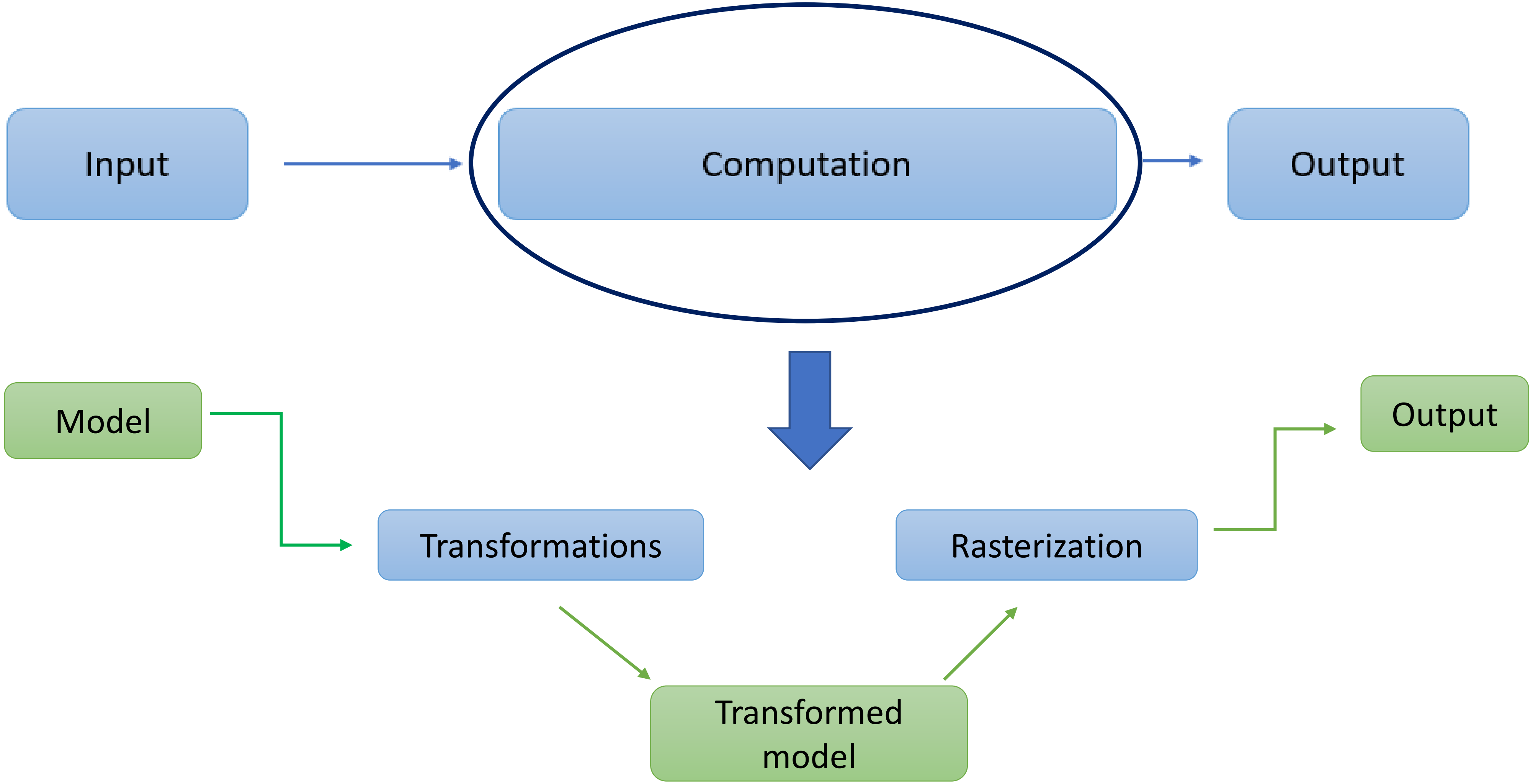
- Image Scanners - Flatbed, etc.
  - What type of data is returned?
    - Bitmap

- Laser Scanners –
  - Emits a laser and does time of flight. Returns 3D point

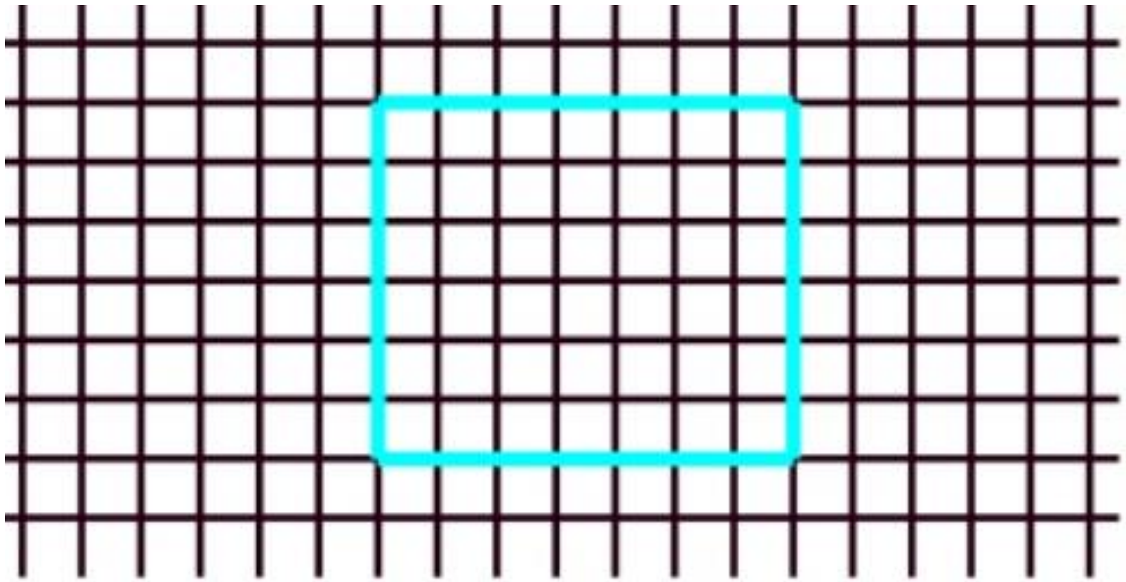


- Camera based – research
  - Examine camera image(s) and try to figure out vertices from them.

- Now we have the model of what we want to draw.



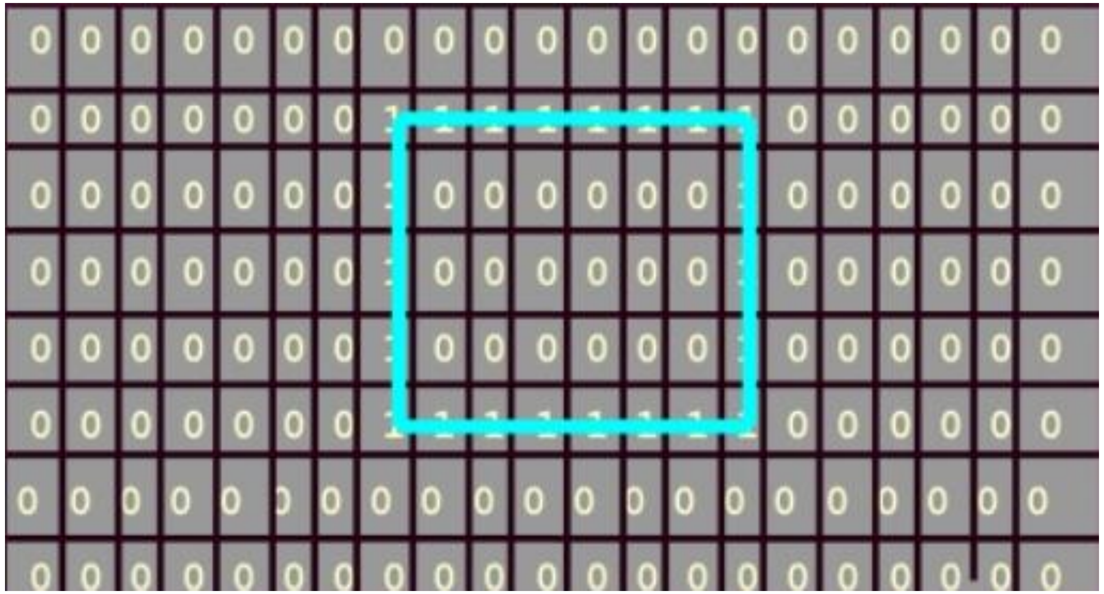
How do we store this ?



- We need to allocate memory to hold the results of the computation stage.

Frame Buffer

- **Framebuffer** - A block of memory, dedicated to graphics output, that holds the contents of what will be displayed.
- **Pixel** - one element of the framebuffer.



Framebuffer in Memory

- If we want a framebuffer of 640 pixels by 480 pixels, we should allocate:

  framebuffer = 640*480 bits (if 1 bit per pixel)

- What would more bits get you ?

  More values to be stored at each pixel

- How many colors does 1 bit get you ?

- How many colors does 8 bit get you ?

Bit Depth

- Bit depth refers to the color information stored in an image
- The higher the bit depth of an image, the more colors it can store.
- 1 bit per pixel
  - can only store one of two values, 0 (white) and 1 (black).
- 8 bits per pixel
  - 256 possible colors
- 16 bits per pixel (high color)
  - 5 bits for red, 5/6 bits for green, 5 bits for blue
  - potential of 32 reds, 32/64 green, 32 blues
  - total colors: 65536
- 32 bits per pixel (true color)
  - 8 bits for red, green, blue, and alpha
  - potential for 256 reds, greens, and blues
  - total colors: 16777216 (more than the eye can distinguish)

Q. What is the Frame Buffer size with 12 bits per pixel For 640 x 480 resolution?

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A. Size = (640*480*12)/8 = 460800 bytes

Q. Suppose a RGB raster system is to be designed using an 8- inch by 10-inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need for the frame buffer?

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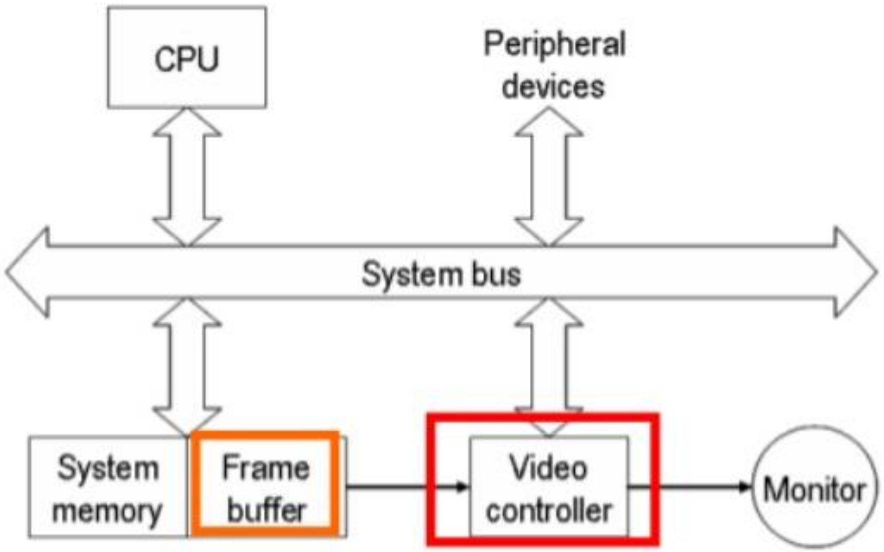
A. Screen Resolution = (8*100)*(10*100)=800*1000 pixels

Storage will be = (800*1000*6)/8 = 600000 bytes

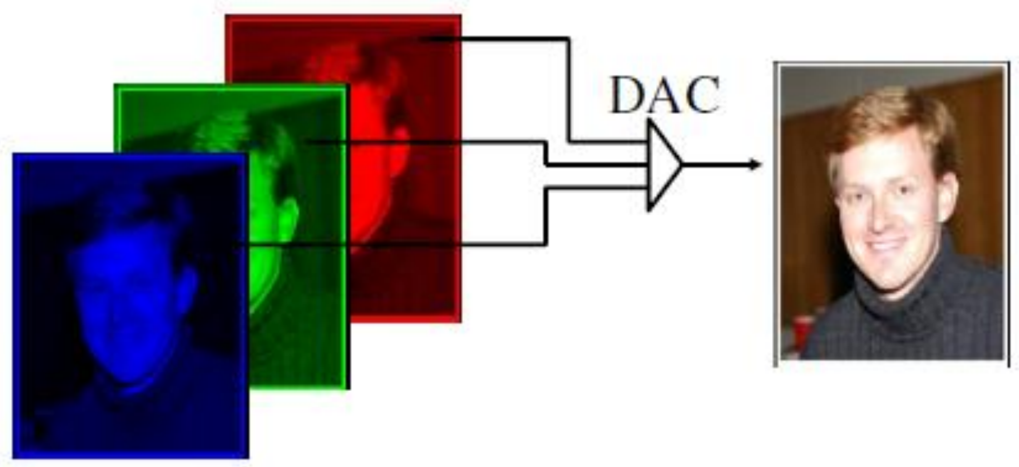
Video Controller

- In Addition to the CPU, a special processor know as Video Controller is used to control the operations of the display system.
- Fixed Area in the system memory is given for the frame buffer, and the video controller is direct access to the framebuffer.



Direct Color Framebuffer

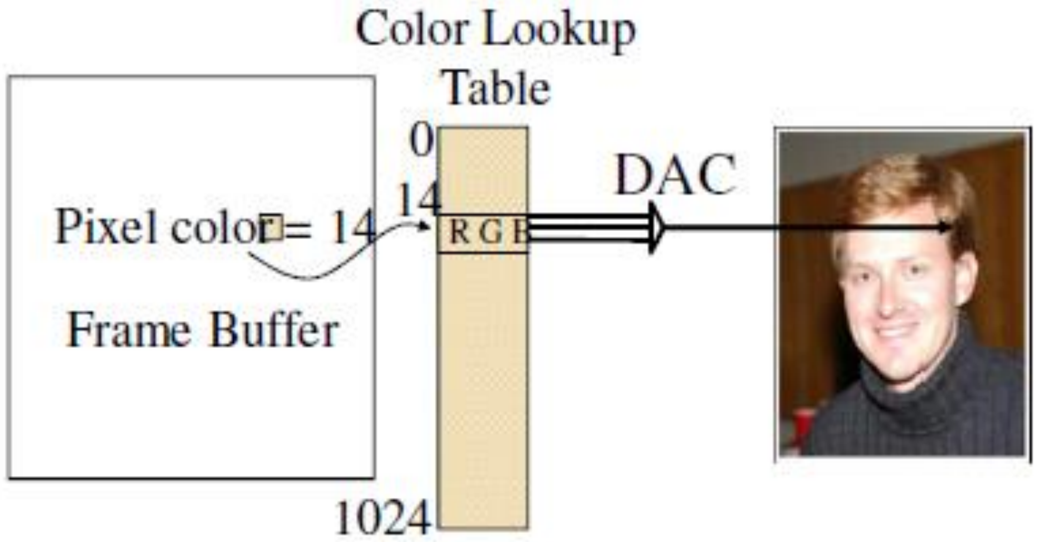
- Store the actual intensities of R, G, and B individually in the framebuffer.
- 24 bits per pixel are commonly used , where 8 bits represent 256 levels of each color.
- If 16-bit color depth is used, usually either use 5 bits per primary color, or use 5 bits for each of red and blue, and 6 bits for green. If 5 bits are used for each color, the 16th bit may either be unused or else used for transparency.



Problem :
It is necessary to read 24-bits for each pixel from frame buffer. This is very time consuming.

Color Lookup Framebuffer

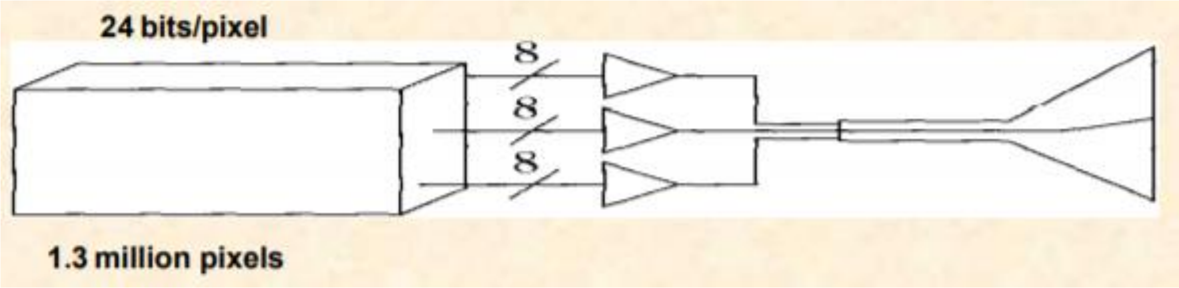
- Used to reduce the frame buffer storage requirement.
- Store indices (usually 8 bits) in framebuffer.
- Display controller looks up the R,G,B values before triggering the electron guns.



Let’s Compare the Cost of two systems

- Suppose two systems with display with l,024-by-l,280-pixel (so that each of them supports about 1.3 million pixels) is there and Both systems allow colors to be defined with a precision of 24 bits, often called "true color.“
- 1 st Approach -> 1024 * 1280 * 24 = ~ 4MByte
  (frame buffer needs ~ 4Mbyte)

Expensive



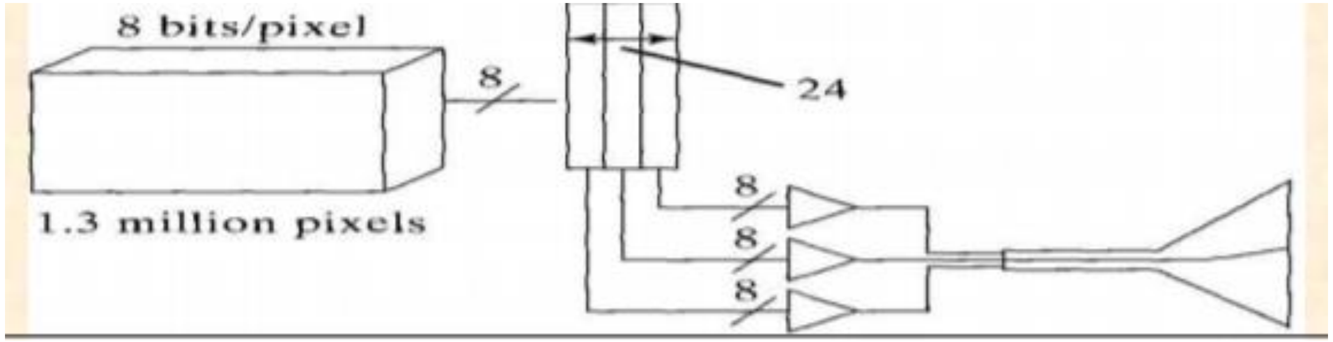
- 2nd Approach ->

  1024 * 1280 * 8 = ~ 1MByte

  LUT requires 256 * 3 = 768 Byte

  (frame buffer needs ~ IMbyte)

Inexpensive



- Now we have an image (model is the framebuffer), now we want to show it.

- Hardcopy
- Display
  - Vector / Random Scan
  - Raster Scan

Q. What are image quality issues that may arise at the output?

Hardcopy

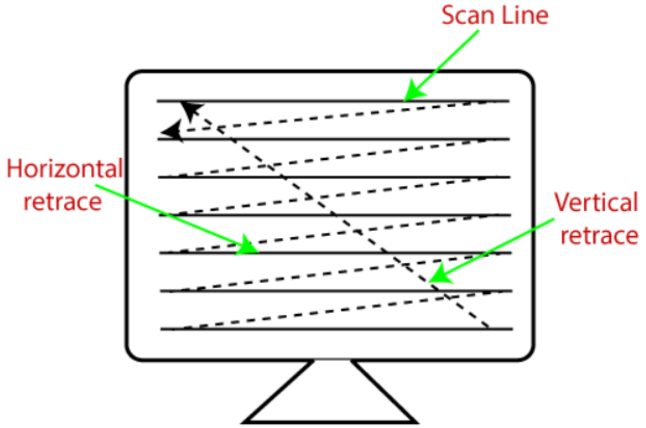
- Printers (Resolution, color depth)
  - Dot Matrix - uses a head with 7 to 24 pins to strike a ribbon (single or multiple color)
  - Ink Jet Printers (fires small balls of colored ink)
  - Laser Printers (powder adheres to positive charged paper)
  - Pen Plotters (similar to vector displays). “infinite” resolution.

Raster Display

- Raster displays typically have an array of addressable dots, which can be individually set to a particular color or intensity.
- In Raster scan type the electron beam sweeps the entire screen from left to right, top to bottom, in the same fashion as we write on a notebook, word by word.
- Raster displays can be implemented by several technologies.
- Currently the most popular is the cathode ray tube (CRT), which can implement both raster and vector displays.
- Liquid Crystal Displays (LCDs) also divided into pixels.

Beam Movement

- **Scan line** - one row on the screen.
- **Interlace vs. Non-Interlace** - Each frame is either drawn entirely, or as two consecutively drawn fields that alternate horizontal scan lines.
- **Horizontal retrace** - The return to the left of the screen after refreshing each scan line is called as the horizontal retrace.
- **Vertical Retrace** - At the end of each frame the electron beam returns to the top left corner of the screen to the beginning the next frame.
- **Refresh rate** - how many frames are drawn per second. Eye can see 24 frames per second. TV is 30 Hz, monitors are at least 60 Hz.



Q. Why Refresh Rates Matter ?

Q. How much time is spent scanning across each row of pixels during scan refresh on a raster system with a resolution of 1280* 1024 and a refresh rate of 60 frames/second ?

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The scan rate for each pixel row is,

60fps *1024 lines/frame = 61,440 lines/sec

Scan time per scan line is,

~ 16.3 microseconds.

(Scan time per frame is 1/60 sec, or approximately 16.7 milliseconds.)

Random Scan Display

- The electron beam is directed only to the part of the screen where the picture is to be drawn .It is also called stroke-writing display, or calligraphic display.
- Picture is stored as a set of points and line drawing commands with (x,y) or (x,y,z) coordinates as well as character plotting commands and stored in an area of memory referred to as refresh display file.
- Pros :
  - Diagrams/Only draw what you need
  - It requires less memory
  - It has very high resolution and limited only by the monitor
- Cons:
  - No fill Objects with complexity
  - Requires processor-controlled beam
  - More Cost

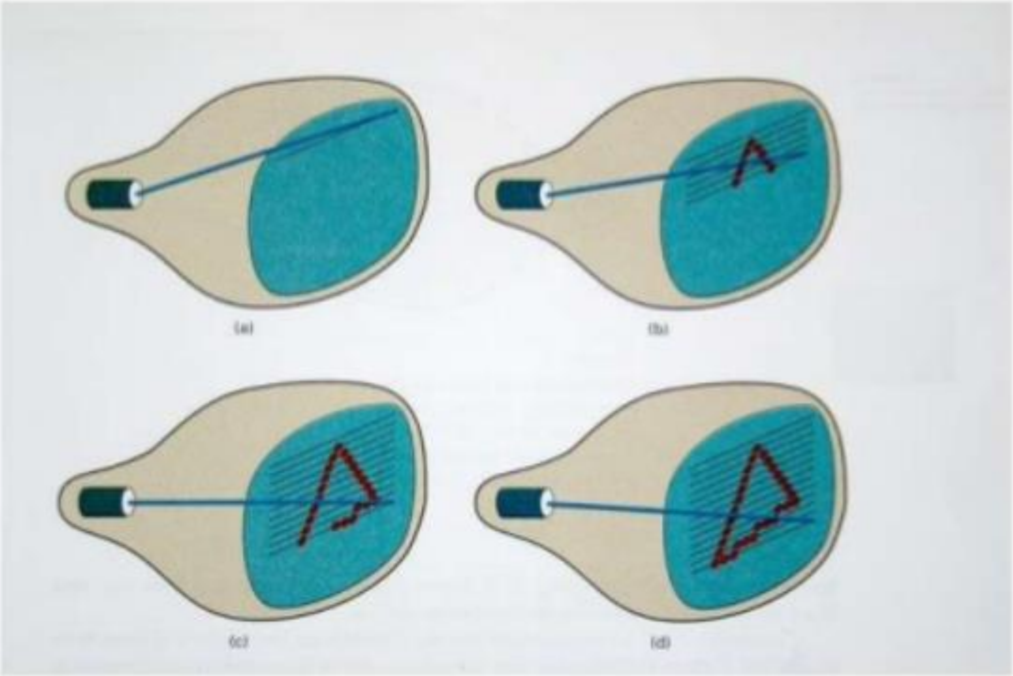


Figure: Drawing a triangle on a Raster Scan Display

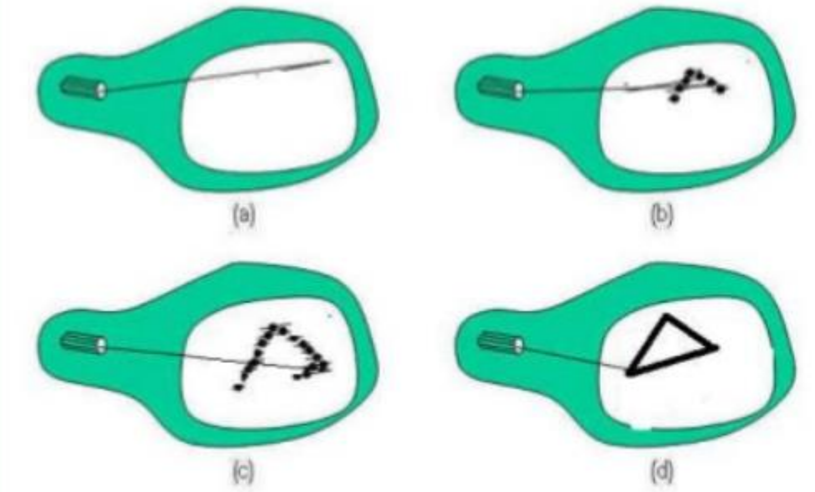
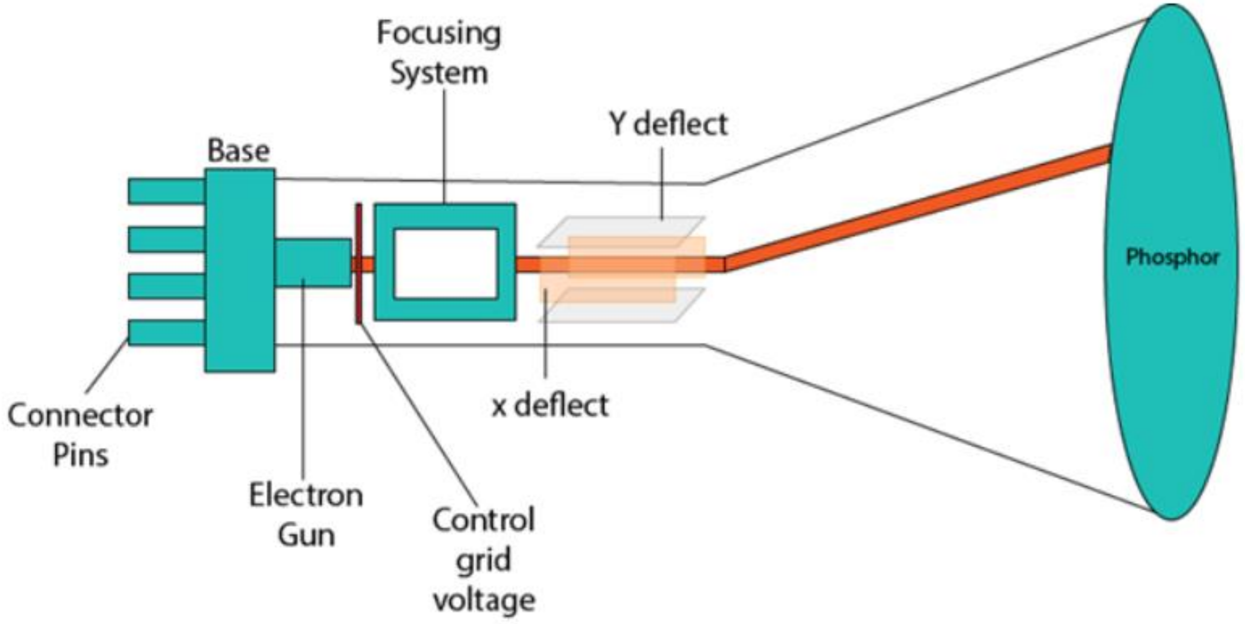


Figure :Drawing a triangle using Random Scan Display

Cathode Ray Tubes (CRTs)

- Standard computer monitor (or TV screen) is based on the cathode ray tube (CRT).



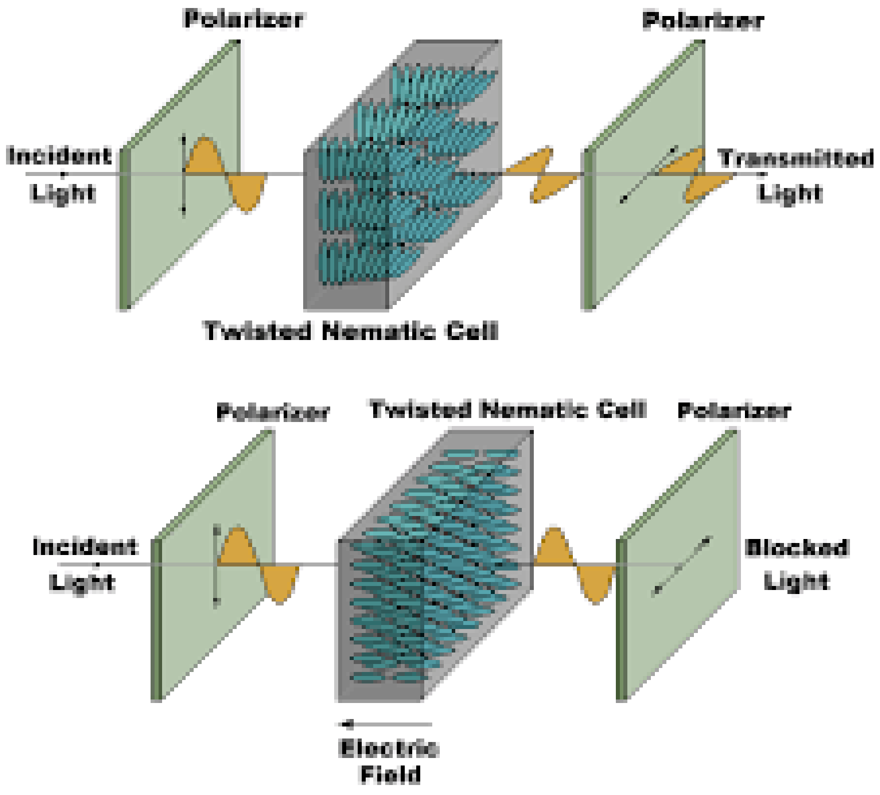
Cathode Ray Tube

- Heating element on the *yolk.*
- *Phosphor* coated screen.
- Electron gun generates a beam of electrons which passes through focusing and deflection system and strikes on specified position on phosphor coated screen.

Liquid Crystal Displays (LCDs)

- Also divided into pixels, but without an electron gun firing at a screen, LCDs have cells that either allow light to flow through or block it.
- Liquid crystal displays use small flat chips which change their transparency properties when a voltage is applied.

Q. What are the Advantages of LCD ?



Questions?